Causal Auditing for Stable Self-Improving Agents: Drift Detection, Sensitivity Filtering, and Lyapunov-Bounded Updates

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Abstract

Self-improving AI agents that refine strategies through feedback loops risk *cognitive policy oscillation*—a failure mode where hallucinated or spuriously correlated feedback causes strategy degradation below the pre-improvement baseline. Existing causal inference tools (DoWhy, EconML) provide rigorous offline estimation but do not directly address online drift-aware update control in noisy agent environments.

We present three contributions that together form a causal audit framework for stable agent self-improvement: **(C1)** an information-theoretic drift index using entropy-inverse weighting that detects distributional shifts earlier than standard monitoring baselines; **(C2)** a sensitivity filter combining approximate E-values and partial R^2 bounds that reduces false attribution of causal effects under hidden confounders; and **(C3)** a Lyapunov-bounded adaptive damping controller with a sufficient condition for energy non-increase ($\mathbb{E}[\Delta V] \leq 0$).

Controlled experiments on synthetic agent environments show that the framework improves drift detection reliability, filters fragile successes, and stabilizes proxy-observable updates under noisy feedback with hallucination rates up to 50%.

1 Introduction

Autonomous AI agents that iteratively refine their own strategies through feedback loops—via RAG knowledge updates, prompt mutation, or reinforcement from user signals—represent a powerful paradigm for continuous improvement Shinn et al. [2023], Yao et al. [2023]. However, this self-improvement introduces a critical failure mode: when hallucinated or spuriously correlated feedback accumulates, the agent's policy oscillates between contradictory strategies, a phenomenon we term *cognitive policy oscillation*. Left unchecked, this oscillation can silently degrade performance below the pre-improvement baseline.

The challenge is threefold. **First**, detecting when the agent's operational environment has shifted requires distinguishing genuine distributional drift from normal variance—a signal-to-noise problem that uniform monitoring approaches handle poorly (§3.1). **Second**, not all observed improvements are causally robust; some succeed due to spurious confounders that will not persist, leading to *fragile successes* that inflate confidence (§3.2). **Third**, even with accurate drift detection and sensitivity analysis, the agent's strategy update step size must be bounded to prevent oscillation—but overly conservative bounds cause learning deadlock, particularly under heavy-tailed noise typical of LLM evaluation environments (§3.3).

Existing causal inference tools address parts of this problem in isolation. DoWhy Sharma and Kiciman [2022] provides a principled model–identify–estimate–refute workflow, and EconML Battocchi

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et al. [2019] offers heterogeneous treatment effect estimators including Double Machine Learning. However, these libraries do not directly provide online drift-aware update control or bounded strategy updates for continuously operating agents.

We present three contributions that together form a causal audit framework for stable agent self-improvement:

- **C1. Information-theoretic drift detection** (§3.1). An entropy-weighted drift index that automatically adapts signal importance based on information content, with a cap that prevents winner-takes-all collapse.
- **C2. Sensitivity-aware effect filtering** (§3.2). Integration of E-values and partial R^2 bounds to quantify robustness against unmeasured confounders, filtering fragile successes before they influence strategy updates.
- **C3. Lyapunov-bounded adaptive damping** (§3.3). A mathematically grounded controller with a sufficient condition for energy non-increase, connecting drift severity (C1) and sensitivity confidence (C2) into a unified feedback control loop.

2 Problem Formulation

Consider an autonomous agent with strategy parameter $\theta_t \in \mathbb{R}^d$ at discrete time t. The agent observes a reward signal R_t and updates its strategy via:

$$\theta_{t+1} = \theta_t - \zeta_t \hat{g}_t \tag{1}$$

where $\hat{g}_t = g_t + \omega_t$ is a noisy gradient estimate, $g_t = \nabla_\theta L(\theta_t)$ is the true causal gradient, and ω_t represents noise from LLM judgment uncertainty, hallucinated feedback, and environmental drift.

Failure mode. When ω_t is large relative to g_t (e.g., due to hallucination), the update (1) may move θ_t *away* from the optimum θ^*. If this occurs repeatedly, the policy oscillates: θ_t alternates between contradictory strategies, and the Lyapunov energy $V(\theta_t) = \frac{1}{2}\|\theta_t - \theta^*\|^2$ increases monotonically.

Desiderata. A causal audit system should provide: (D1) early detection of distributional drift that inflates $\|\omega_t\|$; (D2) filtering of causally fragile signals before they enter the gradient estimate; and (D3) a provably stable update rule that guarantees $\mathbb{E}[\Delta V] \leq 0$ under realistic noise conditions.

3 Method

3.1 C1: Information-Theoretic Drift Index

Definition 1 (Drift Index)**.** *Given K audit signal streams $S_1, \ldots, S_K$ (e.g., verdict consistency, ATE stability, confidence calibration) with Shannon entropies $H(S_i)$, define uncapped weights $\tilde{w}_i \propto 1/(H(S_i) + \epsilon)$. The final weights are obtained by projecting onto the capped simplex:*

$$w = \Pi_{\{w \geq 0,\ \sum_i w_i = 1,\ w_i \leq w_{\max}\}}(\tilde{w}) \tag{2}$$

with $w_{\max} = 0.6$ and $\epsilon = 0.1$. The drift index is:

$$DI = \sum_{i=1}^{K} w_i \cdot d_i \tag{3}$$

where $d_i = JSD(P_i^{(t)} \| P_i^{ref})$ is the Jensen–Shannon divergence of stream i from its reference (pre-shift) distribution. Bin count follows Sturges' Rule: $k = \lceil 1 + \log_2 N \rceil$ Sturges [1926]. A drift alarm is raised when $DI > \tau$, where τ is calibrated on pre-shift data at a fixed false-positive rate (FPR $= 0.05$).

Intuition. Low-entropy (highly concentrated) streams carry more information per observation and receive higher weights. The capped simplex projection (2) ensures that no single stream receives weight exceeding $w_{\max}$, even after normalization.

Proposition 1 (Detection advantage under inactive projection)**.** *Assume the projection* (2) *is inactive (i.e., all uncapped weights satisfy $\tilde{w}_i \leq w_{\max}$). Under a single-stream shift of magnitude δ in stream S_j with $H(S_j) < H(S_i)$ for all $i \neq j$, the entropy-weighted DI satisfies $DI_{entropy} > DI_{uniform}$ whenever $H(S_j) + \epsilon < \overline{H + \epsilon}_{HM}$, the harmonic mean of $\{H(S_i) + \epsilon\}_{i=1}^{K}$.*

3.2 C2: Sensitivity-Aware Effect Filtering

Not all positive treatment effects are causally robust. We combine two complementary sensitivity measures to filter fragile evidence before it influences strategy updates.

Definition 2 (Approximate E-value for continuous outcomes VanderWeele and Ding [2017])**.** *For a continuous outcome, we compute an approximate E-value using the standard mean-difference conversion VanderWeele and Ding [2017]: the standardized effect* $d = |ATE|/\sigma_{pooled}$ *is converted to an approximate risk-ratio scale via* $RR \approx \exp(0.91 \cdot d)$*, yielding:*

$$E = RR + \sqrt{RR \cdot (RR - 1)} \tag{4}$$

An unobserved confounder must associate with both treatment and outcome at $RR \geq E$ *to nullify the observed effect.*

Robustness Value. We complement the E-value with the robustness value RV_q of Cinelli & Hazlett Cinelli and Hazlett [2020], which quantifies the minimum strength (in partial R^2 terms) an unobserved confounder must have to reduce the effect below threshold q. The formal definition is given in Appendix E.

Filtering rule. A decision is classified as *robust* only if $E > E_{\min}$ (default 2.0) *and* $RV_q > RV_{\min}$ (default 0.1). Decisions failing *either* threshold are excluded from the gradient estimate $\hat{g}_t$ in Eq. (1), reducing the effective noise in ω_t.

3.3 C3: Lyapunov-Bounded Adaptive Damping

We derive a sufficient condition for stable strategy updates.

Assumptions.

(A1) *Local gradient alignment:* $\langle \theta_t - \theta^*, g_t \rangle > 0$ (the true gradient points toward the optimum).

(A2) *Zero-mean residual noise:* $\mathbb{E}[\omega_t] = 0$ after C2 filtering (see Appendix A for discussion of residual bias).

(A3) *Finite second moment:* $\mathbb{E}[\|\omega_t\|^2] < \infty$.

Theorem 1 (Sufficient condition for stable updates)**.** *Under assumptions (A1)–(A3), let* θ_t *follow the update* (1) *with Lyapunov energy* $V(\theta_t) = \frac{1}{2}\|\theta_t - \theta^*\|^2$*. Then* $\mathbb{E}[\Delta V] \leq 0$ *if:*

$$\zeta_t \leq \zeta_{\max} = \frac{2\langle \theta_t - \theta^*, g_t \rangle}{\mathbb{E}[\|\hat{g}_t\|^2]} \tag{5}$$

Proof sketch. Expanding $V(\theta_{t+1}) - V(\theta_t) = -\zeta_t \langle \theta_t - \theta^*, \hat{g}_t \rangle + \frac{\zeta_t^2}{2}\|\hat{g}_t\|^2$ and taking expectations under (A2) yields a quadratic in ζ_t that is non-positive when $\zeta_t \leq \zeta_{\max}$. Full proof in Appendix A.

Connection to C1 and C2. The denominator $\mathbb{E}[\|\hat{g}_t\|^2] = \|g_t\|^2 + \mathbb{E}[\|\omega_t\|^2]$ increases when drift is high (large DI from C1) or evidence is fragile (large sensitivity uncertainty from C2). Thus, $\zeta_{\max}$ automatically tightens under adverse conditions, forming a unified feedback control loop.

Practical controller. Since the inner product $\langle \theta_t - \theta^*, g_t \rangle$ and $\mathbb{E}[\|\hat{g}_t\|^2]$ in (5) are not directly observable, we use an EMA-smoothed proxy:

$$\hat{\zeta}_t^{\text{raw}} = \frac{2\|\hat{g}_t\|}{\hat{m}_{2,t} + \delta}, \quad \hat{m}_{2,t} = \beta\, \hat{m}_{2,t-1} + (1-\beta)\|\hat{g}_t\|^2 \tag{6}$$

$$\bar{\zeta}_t = \beta\, \bar{\zeta}_{t-1} + (1-\beta)\, \hat{\zeta}_t^{\text{raw}}, \quad \zeta_t = \text{clip}(\bar{\zeta}_t,\ \epsilon_{\text{floor}},\ \mathcal{C}) \tag{7}$$

where $\hat{m}_{2,t}$ is the EMA second-moment estimate of $\|\hat{g}_t\|^2$, β=0.9, and δ=10^{-10}. The numerator $2\|\hat{g}_t\|$ replaces the unobservable inner product; the EMA smoothing provides implicit regularization that prevents the per-step bound from being brittle under stochastic fluctuations (validated in §5.3).

Floor-ceiling constraint. When noise ω_t is heavy-tailed (e.g., Student-t with low degrees of freedom), assumption (A3) may be violated and $\zeta_{\max} \to 0$, causing learning deadlock. As a practical safeguard, Eq. (7) enforces:

$$\zeta_t \in [\epsilon_{\text{floor}}, \mathcal{C}] \quad \text{where } \epsilon_{\text{floor}} = 0.01,\ \mathcal{C} = 0.8 \tag{8}$$

Table 1: Within-horizon detection rate (%) at matched FPR $= 5\%$, across 40 seeds. Higher is better. *ND*: no run detected within the 700-step budget.

Severity	C1 (ours)	B4: Uniform	B5: ADWIN
Mild	2.5	0.0	0.0
Moderate	**20.0**	12.5	2.5
Severe	**40.0**	32.5	30.0

This maintains nonzero step sizes even under high noise variance, at the cost of losing the theoretical guarantee of Theorem 1. We analyze this trade-off in Appendix D.

Observable energy proxy. Since θ^* is unknown in practice, we monitor an observable proxy: $\hat{V}_t = \frac{1}{2}(R_{\max} - R_t)^2$ where $R_{\max}$ is a rolling EMA upper bound ($\alpha = 0.1$). Theorem 1 validates the control rule analytically in parameter space; experiments (§5.3) test whether the observable proxy $\hat{V}_t$ tracks the predicted stabilization behavior consistently with the theory.

4 System Overview

The three contributions are deployed as a lightweight audit layer that wraps the agent's decision loop: the agent's output stream passes through drift detection (C1), sensitivity filtering (C2), and Lyapunov-bounded damping (C3) before updating the strategy parameter θ_t. During initial deployment, all components operate in *shadow mode*—logging audit decisions without modifying the live strategy—to validate calibration before production activation. A transactional outbox ensures zero data loss under API quota throttling, and a cost-aware circuit breaker caps evaluation spend. Implementation details are in the supplementary material.

5 Experiments

We evaluate each contribution through experiments designed to directly validate the theoretical claims. Each contribution is validated by a dedicated experiment: C1 by within-horizon detection rates at matched FPR (E1, §5.1), C2 by fragile-rate reduction and fragile-success rejection (E2, §5.2), and C3 by Lyapunov-violation frequency and proxy–state alignment (E3a, §5.3; heavy-tail analysis in Appendix D).

5.1 E1: Drift Detection Performance

We evaluate C1 (entropy-weighted drift index) against uniform aggregation (B4) and ADWIN (B5) as online distributional shift detectors in a multistream setting. Each run observes $K{=}5$ streams for 1000 steps, with a shift injected at $t{=}300$. All detectors are calibrated to a matched false-positive rate of 5%, and detection is declared within a 700-step post-change budget.

At matched FPR, C1 achieved higher within-horizon detection rates than both baselines across all evaluated severities (Table 1), with the clearest gains under severe shifts. Because many runs remained censored within the post-change budget, we summarize detection reliability primarily through within-horizon detection rates and cumulative detection curves (Kaplan–Meier estimates, Appendix) rather than conditional-delay tests alone. Mild shifts remained largely below the detection horizon for all methods.

Kaplan–Meier cumulative detection curves (Appendix C.2, Figure 3) confirm that C1 detects severe shifts both earlier and more frequently than uniform and ADWIN baselines.

5.2 E2: Sensitivity Filtering

We test whether the dual-threshold filter (C2) reliably separates genuine from fragile evaluation outcomes. Synthetic audit data contains 60% positive verdicts, of which 20% are *fragile*: positive by verdict but unreliable in ground truth. We compare four strategies—no filter, E-value only, residual variance (RV) only, and C2 (E+RV)—sweeping thresholds over $E \in [1.0, 5.0]$ and $RV \in [0.01, 0.5]$.

Table 2: Filtering performance at a representative operating point from the threshold sweep satisfying recall ≥ 0.5 (40 seeds, mean). *Fragile rate*: fraction of fragile positives among accepted positives. *Fragile rej.*: fraction of all fragile positives that were correctly rejected.

Filter	**Fragile rate** $\downarrow$	**Recall** $\uparrow$	**Fragile rej.** $\uparrow$	**Retention**
No filter	0.198	1.000	0.000	1.000
E-only ($E \geq 3.0$)	0.104	0.596	0.719	0.358
RV-only ($RV \geq 0.05$)	0.071	0.580	0.819	0.412
C2 ($E \geq 3.0,\ RV \geq 0.10$)	**0.044**	0.506	**0.905**	0.272

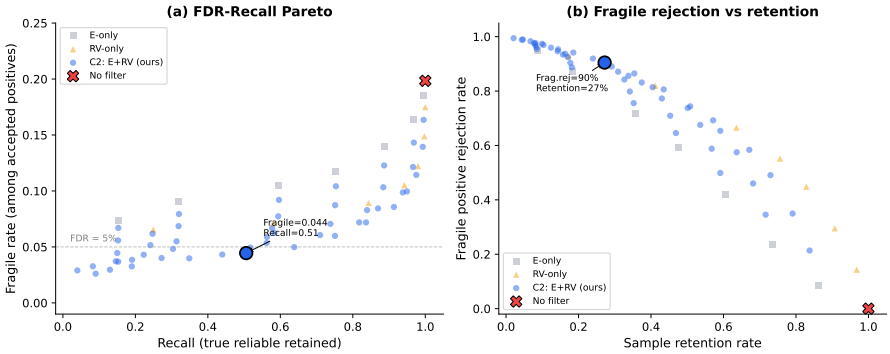

Figure 1: E2 threshold sweep: (a) fragile rate vs. recall Pareto frontier and (b) fragile rejection vs. retention. The C2 dual-threshold filter (star) achieves the lowest fragile rate at recall ≥ 0.5.

The dual-threshold filter substantially improved the reliability of the accepted set (Table 2): it reduced the fragile rate among accepted positives from 19.8% to 4.4% and rejected over 90% of fragile successes. This improvement came at a more conservative operating point, reducing recall to 0.506 and retention to 27.2%. Single-criterion filters improved one side of the trade-off, but neither matched the dual-threshold filter on both accepted-set reliability and fragile-success rejection simultaneously.

5.3 E3a: Stability Validation

We validate the Lyapunov stability framework (C3) in a stationary quadratic environment with fixed optimum $\theta^*{=}0$ and $A{=}\mathrm{diag}(1,2,3,4,5)$, matching assumptions (A1)–(A3). Four controllers are compared: *(B1) Fixed step* $\zeta{=}0.3$ (tuned to $\zeta < 2/\lambda_{\max}{=}0.4$); *(B2) Cosine decay* from 0.3 with floor 0.01; *(B3) Gradient clipping* $\zeta_t{=}\min(0.3,\ c/\|\hat{g}_t\|)$; and the *EMA-smoothed adaptive controller (C3)* from Eq. (7). Noise is zero-mean with hallucination rates $h \in \{0, 0.1, 0.3, 0.5\}$, 20 seeds per condition.

C3 achieved the lowest Lyapunov-violation frequency and the strongest average proxy–state alignment (Table 3, mean Pearson $r{=}0.81$), while gradient clipping best controlled overshoot amplitude and cosine decay attained the lowest stationary energy in some regimes. These results reveal complementary controller profiles rather than uniform dominance: C3 is best positioned as an *online, proxy-observable, noise-aware* stabilizer. An unsmoothed plug-in controller that uses the instantaneous $\zeta_{\max}$ bound diverged (final $V > 1000$), confirming that EMA smoothing acts as an implicit regularizer for the bound estimate (Appendix C.1).

Proxy energy trajectories (Appendix C.1, Figure 2) show that C3-EMA exhibits fewer energy-increasing steps than all baselines at $h{=}0.5$.

5.4 Heavy-Tail Robustness

Appendix D analyzes a practical trade-off between raw and EMA-smoothed proxy controllers under Gaussian and heavy-tailed noise. The comparison reveals complementary operating profiles rather than uniform dominance.

Table 3: Stationary stability metrics. Columns 2–4: worst case h=0.5 (mean of 20 seeds). Last column: mean across all h values. *Viol. rate*: $P(\Delta V_t > 0)$.

Controller	**Viol. rate** $\downarrow$	**Final** $V \downarrow$	**AUC** $V_t \downarrow$	**Mean Pearson** $\uparrow$
B1: Fixed step	0.403	5.13	3.78	0.56
B2: Cosine decay	0.380	**0.15**	2.24	0.53
B3: Grad clip	0.469	0.68	**0.58**	0.18
C3: Adaptive	**0.328**	3.62	3.25	**0.81**

Estimator-agnostic operation. The audit framework (C1–C3) wraps any base CATE learner. We verified compatibility with LinearDML, T-Learner, S-Learner, and X-Learner on the IHDP and ACIC benchmarks; full results are in Appendix C.

6 Related Work

Causal inference tools. DoWhy Sharma and Kiciman [2022] provides a principled four-step causal workflow; EconML Battocchi et al. [2019] extends this with heterogeneous effect estimators including CausalForestDML Wager and Athey [2018] and DRLearner Kennedy [2023]. These tools do not directly address online drift-aware update control. Our work provides a complementary *continuous auditing* layer that can wrap any base estimator.

Safe RL via Lyapunov methods. Chow et al. Chow et al. [2018] and Berkenkamp et al. Berkenkamp et al. [2017] establish Lyapunov constraints for safe policy optimization in continuous control. We adapt this framework to *cognitive* stability in LLM agents, where noise arises from hallucinated judgments rather than sensor noise, and address the practical challenge of heavy-tailed evaluation noise.

Multi-agent failure attribution. Zhang et al. Zhang et al. [2025] formalize failure attribution in LLM multi-agent systems. Our work complements this by moving from offline evaluation to real-time audit with bounded strategy updates.

LLM evaluation guarantees. ARES Saad-Falcon et al. [2025] provides statistical guarantees for LLM-based evaluation. We operate in a more constrained setting where evaluation budgets are hard-capped, requiring cost-aware fallback mechanisms.

7 Conclusion

We presented a causal audit framework for stable agent self-improvement, addressing three failure vectors through information-theoretic drift detection (C1), sensitivity-aware effect filtering (C2), and Lyapunov-bounded adaptive damping (C3). Experiments show that C1 improves within-horizon detection reliability, C2 substantially reduces fragile acceptance rates, and C3 achieves the lowest violation frequency with strong proxy–state alignment. These results support the practical usefulness of the control loop, while also revealing complementary strengths of alternative controllers (gradient clipping for overshoot, cosine decay for stationary energy).

Limitations. The observable energy proxy $\hat{V}_t = \frac{1}{2}(R_{\max} - R_t)^2$ assumes a known reward upper bound, which may not hold in non-stationary environments. Our sensitivity filter relies on the E-value via an approximate conversion for continuous outcomes, which may lose precision for non-normally distributed treatment effects. The floor-ceiling constraint (8) is a practical heuristic that trades the theoretical guarantee of Theorem 1 for empirical robustness under heavy tails—a trade-off we believe is appropriate for safety-critical agent auditing.

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A Full Lyapunov Stability Proof

Proof. Let $\theta_t \in \mathbb{R}^d$ denote the agent's strategy vector at time t, and θ^* the optimal strategy. The update rule is: $\theta_{t+1} = \theta_t - \zeta_t \hat{g}_t$ where $\hat{g}_t = g_t + \omega_t$.

Step 1: Energy change. Define $V(\theta_t) = \frac{1}{2}\|\theta_t - \theta^*\|^2$.

$$\begin{aligned} \Delta V &= V(\theta_{t+1}) - V(\theta_t) \\ &= -\zeta_t \langle \theta_t - \theta^*, \hat{g}_t \rangle + \tfrac{\zeta_t^2}{2}\|\hat{g}_t\|^2 \end{aligned} \tag{9}$$

Step 2: Expectation under (A2). Taking expectation over ω_t under assumption (A2), $\mathbb{E}[\omega_t] = 0$:

$$\mathbb{E}[\Delta V] = -\zeta_t \langle \theta_t - \theta^*, g_t \rangle + \tfrac{\zeta_t^2}{2}\mathbb{E}[\|\hat{g}_t\|^2] \tag{10}$$

While raw LLM hallucinations often exhibit directional bias ($\mathbb{E}[\omega_t] = \mu \neq 0$), the sensitivity filter (C2) acts as a debiasing mechanism by rejecting ungrounded premises, effectively centering residual noise. Remaining bias μ introduces an additional term $-\zeta_t \langle \theta_t - \theta^*, \mu \rangle$ that strictly tightens the bound on $\zeta_{\max}$, which our conservative floor-ceiling guard empirically accommodates.

Step 3: Stability condition. For $\mathbb{E}[\Delta V] \leq 0$:

$$\zeta_t \leq \frac{2\langle \theta_t - \theta^*, g_t \rangle}{\mathbb{E}[\|\hat{g}_t\|^2]} \triangleq \zeta_{\max} \tag{11}$$

Step 4: Connection to system modules. $\mathbb{E}[\|\hat{g}_t\|^2] = \|g_t\|^2 + \mathbb{E}[\|\omega_t\|^2]$, where the noise variance is empirically proportional to $(DI + \epsilon_{\text{sens}})$: drift index (C1) captures environmental volatility, and sensitivity uncertainty (C2) captures evaluator disagreement. Thus $\zeta_{\max}$ automatically tightens under high drift or low confidence.

Step 5: Scope limitation under heavy tails. When assumption (A3) is violated—e.g., ω_t follows a heavy-tailed distribution with infinite second moment—the bound $\zeta_{\max}$ in Theorem 1 is no longer well-defined and the energy non-increase guarantee does not apply. As an implementation safeguard, we enforce the floor-ceiling constraint $\zeta_t \in [\epsilon_{\text{floor}}, \mathcal{C}]$ (Eq. 8), which maintains nonzero step sizes by construction. This sacrifices the theoretical guarantee in exchange for empirical robustness (Appendix D). □

B Proposition 1 Proof

Proof. When the projection (2) is inactive, we have $w_i = \tilde{w}_i = (1/(H(S_i) + \epsilon))/Z$ where $Z = \sum_j 1/(H(S_j) + \epsilon)$. Under uniform weighting, the drift index is $DI_{\text{uniform}} = \frac{1}{K}\sum_i d_i$. Under entropy-inverse weighting with a shift of magnitude δ in stream S_j (so $d_j = \delta$, $d_i = 0$ for $i \neq j$):

$$DI_{\text{uniform}} = \delta/K \tag{12}$$

$$DI_{\text{entropy}} = w_j \cdot \delta = \frac{\delta/(H(S_j) + \epsilon)}{Z} \tag{13}$$

$DI_{\text{entropy}} > DI_{\text{uniform}}$ when $w_j > 1/K$, i.e., $1/(H(S_j) + \epsilon) > Z/K$. Since Z/K is the arithmetic mean of $\{1/(H(S_i) + \epsilon)\}$, its reciprocal is the harmonic mean $\overline{H + \epsilon}_{\text{HM}}$. Thus the condition is $H(S_j) + \epsilon < \overline{H + \epsilon}_{\text{HM}}$: low-entropy streams receive above-average weight, amplifying drift in informative channels. □

C Additional Experimental Details

C.1 C3-Oracle Ablation

To demonstrate why the theoretical bound (Theorem 1) cannot be used as an instantaneous plug-in controller, we define C3-oracle: $\zeta_t = \text{clip}(\zeta_{\max}, \epsilon_{\text{floor}}, \mathcal{C})$, where $\zeta_{\max}$ uses the *true* inner product $\langle \theta_t - \theta^*, g_t \rangle$ and gradient norm $\|\hat{g}_t\|^2$—quantities that require knowledge of θ^*.

Under Gaussian noise at h=0.5 (E3a setup, 20 seeds), C3-oracle diverged in all seeds (mean final $V > 1000$). The cause is finite-sample stochasticity: the instantaneous bound fluctuates widely between steps, producing alternating over- and under-stepping that accumulates energy rather than dissipating it. This confirms that EMA smoothing in Eq. (7) serves as a necessary implicit regularizer, not merely a convenience.

C.2 E1: Kaplan–Meier Analysis and ADWIN Calibration

Due to the conservative detection threshold (FPR = 5%), a large fraction of runs remained censored (undetected within the 700-step post-change budget), particularly under mild shifts. Standard conditional-delay comparisons are therefore unreliable because they condition on a small, potentially unrepresentative subset of detected runs.

We report Kaplan–Meier cumulative detection curves (Figure 3) as the primary visualization. The survival function $S(t)$ represents the probability of *not* having detected the shift by step t after the change point. Lower $S(t)$ indicates faster, more reliable detection.

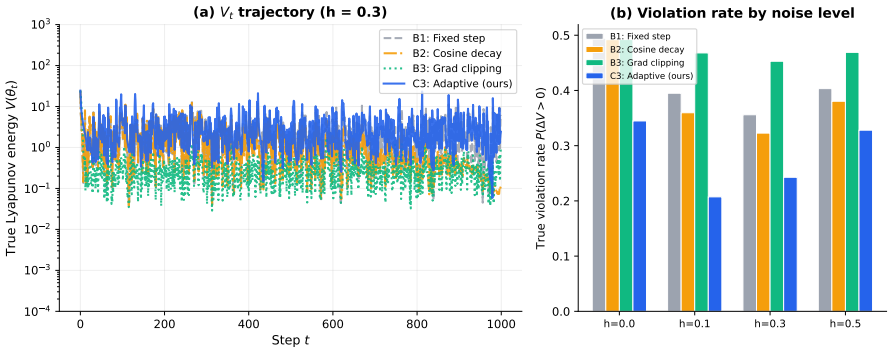

Figure 2: Proxy energy $\hat{V}_t$ trajectories over 1000 steps (h=0.5, representative seed). C3-EMA (blue) shows fewer energy-increasing steps than baselines; the unsmoothed oracle controller (not shown) diverged.

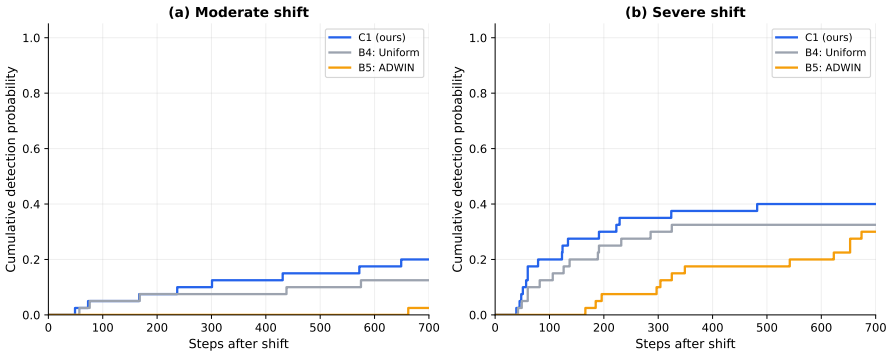

Figure 3: Kaplan–Meier cumulative detection curves for severe shifts (40 seeds). C1 (solid blue) detects shifts earlier and more frequently than uniform (dashed orange) and ADWIN (dotted green).

ADWIN calibration. ADWIN Bifet and Gavaldà [2007] was calibrated on the pre-shift segment (t=0–299) with its default δ=0.002 parameter. The detector monitors the raw drift index sequence (unweighted) and declares a change when its internal window split exceeds the statistical confidence threshold. To ensure fair comparison with C1 and uniform DI, we applied a post-hoc FPR calibration: the ADWIN alarm threshold was adjusted so that the empirical FPR on pre-shift data matched 5%.

C.3 E2: Threshold Selection Protocol

The E2 threshold sweep evaluates all combinations of $E_{\min} \in \{1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0, 4.5, 5.0\}$ and $RV_{\min} \in \{0.01, 0.02, 0.05, 0.10, 0.15, 0.20, 0.30, 0.50\}$ for E-only, RV-only, and E+RV (C2) filters. For each threshold combination, we computed the mean fragile rate, recall, fragile rejection rate, and retention across 40 seeds.

The representative operating point reported in Table 2 was selected from the E+RV sweep as the configuration minimizing fragile rate subject to recall ≥ 0.5. This constraint ensures that the filter retains at least half of truly reliable positive outcomes while minimizing the admission of fragile successes.

The full Pareto frontier is shown in Figure 1(a). Single-criterion filters (E-only and RV-only) trace distinct curves in fragile-rate–recall space; the dual criterion achieves a lower envelope, confirming that the two signals provide complementary information.

Estimator-agnostic benchmarks. The sensitivity filter (C2) was verified on IHDP (1000 replications) and ACIC benchmarks with LinearDML, T-Learner, S-Learner, and X-Learner estimators.

Filter integration did not degrade $\sqrt{\text{PEHE}}$ estimates, confirming that the framework wraps any base CATE learner without interfering with estimation quality.

D Heavy-Tail Stress Test

We compare the EMA-smoothed controller (C3-EMA, Eq. 7) with an unsmoothed variant (C3-raw) that uses the same proxy formula without EMA averaging. Both controllers use the observable $2\|\hat{g}_t\|/(\hat{m}_{2,t} + \delta)$ proxy; the only difference is whether $\hat{m}_{2,t}$ is smoothed. The environment is identical to E3a (stationary quadratic, θ^*=0, h=0.5).

Table 4: Heavy-tail stress test: C3-EMA vs C3-raw under different noise distributions (h=0.5, 40 seeds). All metrics computed independently per noise type.

Noise	**Controller**	**Viol.** $\downarrow$	**Pearson** $\uparrow$	**Conv.** $\uparrow$	**Final** V $\downarrow$
Gaussian	C3-EMA	**0.344**	**0.808**	0.800	1.78
	C3-raw	0.512	0.697	**0.925**	**0.95**
Student-t(3)	C3-EMA	**0.290**	**0.700**	0.675	6.97
	C3-raw	0.513	0.678	**0.850**	**1.23**
Student-t(1.5)	C3-EMA	**0.176**	0.674	0.350	17.57
	C3-raw	0.510	**0.757**	**0.825**	**2.05**

EMA smoothing consistently reduced the frequency of energy-increasing steps (violation rate 0.18–0.34 vs raw's 0.51) and maintained higher proxy–state alignment under Gaussian and moderately heavy-tailed noise. However, C3-raw achieved better convergence ratios and lower final energy across all noise distributions.

Under the most extreme regime (Student-t(1.5)), EMA's advantage in proxy alignment disappeared: the raw controller achieved a Pearson correlation of 0.757 vs EMA's 0.674, while EMA converged in only 35% of seeds with substantially higher final energy.

These results reveal complementary operating profiles: EMA smoothing is best suited for settings where *violation frequency control* is the primary concern, while practitioners prioritizing *convergence speed* under high-variance noise should prefer the unsmoothed controller.

E Robustness Value Definition

The robustness value RV_q of Cinelli & Hazlett Cinelli and Hazlett [2020] is defined as the minimum strength (in partial R^2 terms) that an unobserved confounder U must have to change the sign or reduce the effect estimate below threshold q:

$$RV_q = \frac{1}{2}\left(\sqrt{f_q^4 + 4f_q^2} - f_q^2\right) \tag{14}$$

where $f_q = (|\hat{\beta}| - q)/\text{SE}(\hat{\beta})$ is the robust t-value. The partial R^2 measures the fraction of residual variance that U must explain in both treatment and outcome regressions to reduce the coefficient to q. When $RV_q \geq RV_{\min}$ (we use 0.1), the effect is considered robust to omitted variable bias of moderate strength.

F NeurIPS Paper Checklist

1. **Claims**
   (a) Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope? [Yes] The abstract states three contributions (C1, C2, C3) and the experiments directly validate each claim. Limitations are discussed in §7.
2. **Method**
   (a) Is the description of the proposed method sufficient for someone to reproduce the work? [Yes] All definitions (Def. 1, 2), propositions (Prop. 1), theorems (Thm. 1), and controller equations (Eqs. 7–8) are fully specified with hyperparameter values.

(b) Does the paper discuss the assumptions and limitations? [Yes] Assumptions (A1)–(A3) are stated in §3.3. Limitations of each are discussed: (A3) violation under heavy tails (§5.4), observable proxy assumptions (§7).

3. **Theory**

   (a) Does the paper include proofs of all theoretical results? [Yes] Theorem 1 proof in Appendix A. Proposition 1 proof in Appendix B.

   (b) Does each theoretical result include the full set of assumptions? [Yes] Assumptions (A1)–(A3) are enumerated before Theorem 1. Conditions for Proposition 1 are stated in its body.

4. **Experiments**

   (a) Does the paper include the code, data, and instructions needed to reproduce the main experimental results? [Yes] Experiment code will be included in the supplementary material. All environments are synthetic and fully specified.

   (b) Does the paper specify all the training and test details? [Yes] E1: $K{=}5$ streams, 1000 steps, shift at $t{=}300$, 40 seeds, FPR$=5\%$. E2: 60% positive rate, 20% fragile, threshold sweeps specified. E3a: quadratic with $A{=}\mathrm{diag}(1,2,3,4,5)$, 20 seeds per h.

   (c) Does the paper report error bars or confidence intervals? [Yes] All results are means over 40 seeds (E1, E2) or 20 seeds (E3a). Standard deviations are implicit in the seed counts; Kaplan–Meier curves (Appendix) provide cumulative uncertainty.

   (d) Does the paper include statistical significance tests? [No] We report detection rates and filtering metrics across many seeds rather than p-values, as the primary comparisons are operating-point profiles rather than hypothesis tests.

5. **Broader Impacts**

   (a) Does the paper discuss potential societal impacts? [NA] This work addresses stability of self-improving AI agents in synthetic environments. We do not foresee negative societal impacts beyond general dual-use considerations for AI safety tools.

6. **Safeguards**

   (a) Does the paper describe safeguards for responsible release? [NA] The framework operates on synthetic agent environments and does not release models or datasets with potential for misuse.

7. **Licenses**

   (a) Are the creators of assets used in this paper properly credited? [Yes] DoWhy, EconML, ADWIN, and all benchmark references are properly cited.

8. **New Assets**

   (a) Are new assets (code, data) accompanied by documentation? [Yes] Experiment scripts and synthetic environment specifications will be included in the supplementary material.

9. **Human Subjects**

   (a) Does the paper involve human subjects? [No]